

FHL ASSISTIVE PROTOTYPE

FARO

Familiar-space Awareness, Recognition, and Orientation

Remembering places. Recognizing hazards. Co-designed with lived experience.

AUTHORS

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Native iPhone app + private vision service + ESP32 module



Familiar spaces can still be disorienting

THE PERSONAL QUESTION

“Where am I in my own environment?”

Generic object recognition cannot answer this by itself.



Orientation

Entering the wrong room or losing track of location.



Context

Understanding nearby objects without a visually demanding interface.



Obstacles

Noticing furniture or another object in the forward path.

FARO is a secondary assistive companion—not a replacement for a white cane, guide dog, or mobility training.

One companion, three distinct jobs



PERSONAL MEMORY

Remember

Learn user-named places from several camera views, then recognize them on-device—or report uncertainty.



REQUESTED CONTEXT

Describe

Send one user-requested still to a private vision service and speak a concise bilingual description.



LOCAL REFLEX

Warn

Measure forward distance on the ESP32 and drive passive-buzzer tones without waiting for cloud AI.

 Large controls, VoiceOver, English and Spanish, touch and voice access

Intentional by design

⏸ DEFAULT AT LAUNCH

Inactive

- Local proximity tones stay silent
- Navigation-oriented output is gated
- Mode is never persisted by the iPhone

🚶 EXPLICITLY ARMED

Navigating

- Arms the ESP32 sonar-to-buzzer path
- Enables “Where am I?” and “What is ahead?”
- Distinct confirmation tone marks transition

Remember



Navigate



Recognize



Describe

Three components, two paths



Native iPhone app

Camera, personal place memory, speech, language, mode control, and BLE orchestration.

AVFoundation

Vision

SwiftData

Speech



Private vision service

Authenticated still-image request → concise description in the selected language.



ESP32 module

Forward sonar, local passive buzzer, and versioned BLE telemetry.

Advisory: camera + context → iPhone → speech

Immediate: sonar → ESP32 → buzzer

The immediate warning never waits

THE CRITICAL INVARIANT

 **sonar** → **ESP32** → **passive buzzer**



Bluetooth lost



App unavailable



Network down



AI unavailable

Once Navigating is armed, the local alert continues.

~~sonar~~ → ~~iPhone~~ → ~~AI~~ → ~~warning~~

A reset or power cycle always returns the ESP32 to silent Inactive mode.

Accessible in English and Spanish

SIX COMMANDS, TWO LANGUAGES

“Describe” · “Describe”

“Where am I?” · “¿Dónde estoy?”

“Remember this as...” ·
“Recuerda este lugar como...”

“What is ahead?” · “¿Qué hay
delante?”

“Start navigation” · “Inicia
navegación”

“Stop navigation” · “Detén
navegación”

 VoiceOver

 English + Spanish

 Large touch targets

OPT-IN HANDS-FREE

“Hey FARO”

“Hola FARO”

Siri opens FARO first. One wake phrase opens one bounded command window.

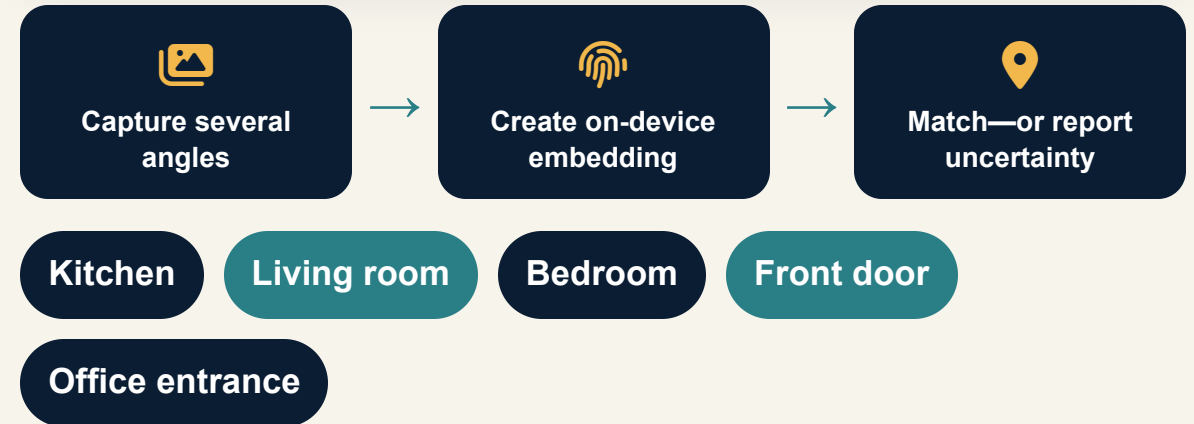
Honest constraints

- Foreground only
- On-device recognition
- Manual Start/Finish fallback
- FARO suppresses its own speech

A seven-step demonstration

- 1 **Start safe:** boot Inactive and silent.
- 2 **Choose access:** switch language or use voice.
- 3 **Teach:** “Remember this as Kitchen.”
- 4 **Recognize:** return from another angle.
- 5 **Describe:** request nearby context.
- 6 **Warn:** move an obstacle through sonar bands.
- 7 **Stop:** leave Navigating and confirm silence.

GUIDED PLACE ENROLLMENT



Implemented and physically integrated

117

iOS tests passed

168

Parameterized test runs

0

Test failures

📷 Physical iPhone camera and spoken description

📄 Authenticated live backend requests

📍 Place enrollment, storage, and uncertainty

🎤 Manual and foreground Hands-Free voice

📶 Automatic iPhone-to-ESP32 connection

🔊 Boot-to-Inactive and local buzzer behavior

✓ Phases 0–6 complete · device-facing gates physically accepted

Clear boundaries are part of the design



One forward sonar

Can miss thin, soft, angled, high or low obstacles, stairs, and drop-offs.



Advisory AI

Scene narration can be wrong or unavailable and never controls the local warning.



Coarse location

GPS may filter different sites; it is never presented as room-level evidence.



Foreground wake

Siri or another supported action must first bring FARO to the foreground.



Recognition uncertainty

Large viewpoint or lighting changes can reduce confidence; FARO must not force a guess.



Prototype hardware

Distance bands are starting points, and the printable holder still needs fit validation.



FARO

Remembering places. Recognizing hazards.

Personalized

Places are named and learned by the user.

Local where it matters

The immediate warning stays on the ESP32.

Explicit about uncertainty

FARO reports limits instead of pretending certainty.



Co-designed with lived experience